

SANJAY NANA LAKSHMANAN

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PROFESSIONAL SUMMARY

Ph.D. Mechanical Engineer with 9 years of combined R&D and academic experience in structural mechanics, thermal and multiphysics analysis. Deep expertise in evaluating consumer electronics component modules, specifically OLED displays, Compute PCBs, and high-density packaging. Proven ability to translate complex boundary value problems such as nonlinear contact mechanics and thermo-mechanical stress into robust design solutions. Adept at cross-functional collaboration and delivering insights for product design.

SKILLS

Mechanical Design: 3D CAD Modeling (SolidWorks, SpaceClaim, Design Modeler), GD&T.

Reliability Physics: Thermal Cycling, Coffin-Manson (Low Cycle Fatigue), Solder Joint Reliability.

Structural Analysis (FEA): Stress Analysis, Contact Mechanics, Nonlinear Material Modeling, Random Vibration.

Simulation Tools: ANSYS Mechanical, ABAQUS, ANSYS Icepak, Flotherm, ANSYS Fluent, STAR-CCM+, COMSOL.

Scripting & Automation: Python (SciPy/Pandas), APDL, C++, FORTRAN (User Subroutines).

Data-Driven Engineering: Design of Experiments (DOE), Surrogate Modeling, Reduced-order modeling.

Communication: Presentations to diverse stakeholders, technical writing and journal publications.

Project Management: Cross-functional collaboration, team leadership/mentorship, multi-tasking.

EDUCATION

Ph.D. in Mechanical Engineering, University of Rochester, Rochester, NY **2016-2022**

M.S. in Mechanical Engineering, University of Rochester, Rochester, NY **2012-2014**

B.E. in Mechanical Engineering, B.M.S College of Engineering, Bengaluru, India **2008-2012**

PROFESSIONAL EXPERIENCE

R&D VERIFICATION ENGINEER

Aug 2022 – May 2025

ANSYS | Canonsburg, PA

- **Reliability Physics:** Validated User Programmable Features (UPF) for stress migration and plastic heat generation, eliminating numerical artifacts in electromigration predictions and improving solver reliability for semiconductor packaging workflows.
- **Thermal Management:** Benchmarked kinematic view factor updates against analytical solutions, resolving convergence instabilities in radiation solvers and enabling high-fidelity thermal prediction for complex systems.

GRADUATE RESEARCH ASSISTANT

Jan 2016 – Feb 2022

University of Rochester | Rochester, NY

- **Wafer/Optic Warpage Prediction (Twyman Effect):** Quantified grinding-induced deflections in thin components using Finite Element simulations. Correlated non-linear geometrical changes and residual stresses to predict curvature, directly applicable to mitigating wafer bow and die warpage in semiconductor fabrication.
- **Contact Mechanics & Adhesion:** Simulated complex surface-surface adhesion behavior between polymer stamps and rigid substrates using a nonlinear traction-separation law. Correlated simulation results with experimental pull-off force values within 3% accuracy effectively separating interface properties from bulk behavior.

PROJECT ASSISTANT

Jun 2014 - Dec 2015

University of Rochester | Rochester, NY

- **Diffusion & Material Modeling:** Developed a Deformation Mechanism Map (DMM) to investigate plastic deformation behaviors, including creep and diffusion, over a range of stresses and temperatures.

SELECTED TECHNICAL PROJECTS

High-Vacuum Process Modeling (GLAD)

- Simulated localized melting and thermal transport in nanostructured silica (GLAD) waveplates processed in a high-vacuum chamber ($<2.5 \times 10^{-5}$ Torr).
- Balanced analytical and computational approaches by first developing a 1D analytical model to understand thermal property effects, followed by a rigorous 2D axisymmetric numerical model in ANSYS.
- Validated that processing must occur under high vacuum to prevent void fraction trapping. Correlated temperature-dependent material properties (porosity, thermal conductivity) to establish a predictive direct-write protocol for precision optics manufacturing.

OLED Micro-Transfer Printing & Display Module Assembly (Ph.D. Research)

- Mechanical Design & Validation: Investigated the micromechanics of contact printing for delicate OLED screens and display components.
- FEA & Contact Mechanics: Utilized targeted plane stress models to analyze complex buckling problems and evaluate pattern collapse mechanisms during component transfer.
- Manufacturing Impact: Correlated FEA adhesion models with empirical pull-off forces within 3% accuracy, effectively predicting component survivability and providing DFM insights for high-volume display manufacturing.

BGA Thermo-Mechanical Reliability & Fatigue Life Simulation

- Simulated complex thermal power cycling on a Ball Grid Array (BGA) package to evaluate thermo-mechanical stress, material behavior, and boundary non-linearities in solder joints.
- Applied the Coffin-Manson empirical relationship to finite element analysis (FEA) outcomes to estimate low-cycle fatigue life based on plastic strain ranges.
- Deepened practical understanding of Design for Reliability (DFR) principles by analyzing how cyclic thermal loads impact the long-term structural integrity of electronic packaging.

CERTIFICATIONS

[Introduction to Computational Fluid Dynamics](#) | ASME

[Probability Foundations for Data Science and AI](#) | Coursera

LEADERSHIP EXPERIENCE

ANSYS

Aug 2022 - May 2025

- Mentored 5 junior engineers and peers on best practices for APDL scripting and Git version control significantly improving team proficiency in automating simulation workflows.

UNIVERSITY OF ROCHESTER

- Graduate Teaching Assistant: Continuum Mechanics (ME 444) Aug 2017 - Dec 2017
- Graduate Teaching Assistant: Finite Elements (ME 441) Aug 2016 - Dec 2016

PEER REVIEWED PUBLICATIONS

- Marcela Mireles, Brittany Hoffman... S. Lakshmanan... et al, [Direct-write laser assisted patterning of form birefringence in waveplates fabricated by glancing angle deposition](#), Optica (2023)
- S. Kim, S. Lakshmanan et al, [Modulation of Interfacial Adhesion Using Semicrystalline Shape-Memory](#), Langmuir (2022)
- S. Lakshmanan and J. Lambropoulos, [Twyman effect in thin curved optics: effects of variable thickness and curvature](#), Appl. Optics (2021)